

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
```

```
In [2]: df=pd.read_csv('heart.csv')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	



```
In [5]: df.info
```

```
Out[5]: <bound method DataFrame.info of
thalach  exang  oldpeak  \
0      52    1    0      125  212    0      1      168    0      1.0
1      53    1    0      140  203    1      0      155    1      3.1
2      70    1    0      145  174    0      1      125    1      2.6
3      61    1    0      148  203    0      1      161    0      0.0
4      62    0    0      138  294    1      1      106    0      1.9
...     ...    ...    ..      ...    ...    ...      ...      ...      ...      ...
1020    59    1    1      140  221    0      1      164    1      0.0
1021    60    1    0      125  258    0      0      141    1      2.8
1022    47    1    0      110  275    0      0      118    1      1.0
1023    50    0    0      110  254    0      0      159    0      0.0
1024    54    1    0      120  188    0      1      113    0      1.4

      slope  ca  thal  target
0         2   2    3        0
1         0   0    3        0
2         0   0    3        0
3         2   1    3        0
4         1   3    2        0
...     ...  ..    ...      ...
1020     2   0    2        1
1021     1   1    3        0
1022     1   1    2        0
1023     2   0    2        1
1024     1   1    3        0

[1025 rows x 14 columns]>
```

```
In [6]: df.describe()
```

Out[6]:

	age	sex	cp	trestbps	chol	fbs	rest
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	54.434146	0.695610	0.942439	131.611707	246.000000	0.149268	0.500000
std	9.072290	0.460373	1.029641	17.516718	51.59251	0.356527	0.500000
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	48.000000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000
50%	56.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	275.000000	0.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

In [8]: df.shape

Out[8]: (1025, 14)

In [21]: df1=df[['age','sex','cp']]
df1

Out[21]:

	age	sex	cp
0	52	1	0
1	53	1	0
2	70	1	0
3	61	1	0
4	62	0	0
...	...	...	...
1020	59	1	1
1021	60	1	0
1022	47	1	0
1023	50	0	0
1024	54	1	0

1025 rows × 3 columns

In [22]: df2=df[['cp','trestbps','chol','fbs']]
df2

Out[22]:

	cp	trestbps	chol	fbs
0	0	125	212	0
1	0	140	203	1
2	0	145	174	0
3	0	148	203	0
4	0	138	294	1
...	...	...	...	...
1020	1	140	221	0
1021	0	125	258	0
1022	0	110	275	0
1023	0	110	254	0
1024	0	120	188	0

1025 rows × 4 columns

```
In [23]: df3=df[['restecg','thalach','exang','oldpeak']]
df3
```

Out[23]:

	restecg	thalach	exang	oldpeak
0	1	168	0	1.0
1	0	155	1	3.1
2	1	125	1	2.6
3	1	161	0	0.0
4	1	106	0	1.9
...	...	...	...	...
1020	1	164	1	0.0
1021	0	141	1	2.8
1022	0	118	1	1.0
1023	0	159	0	0.0
1024	1	113	0	1.4

1025 rows × 4 columns

```
In [24]: merg=pd.concat([df1,df2,df3])
merg
```

Out[24]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
0	52.0	1.0	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	53.0	1.0	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	70.0	1.0	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	61.0	1.0	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	62.0	0.0	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...
1020	NaN	NaN	NaN	NaN	NaN	NaN	1.0	164.0	1.0	0.0
1021	NaN	NaN	NaN	NaN	NaN	NaN	0.0	141.0	1.0	2.8
1022	NaN	NaN	NaN	NaN	NaN	NaN	0.0	118.0	1.0	1.0
1023	NaN	NaN	NaN	NaN	NaN	NaN	0.0	159.0	0.0	0.0
1024	NaN	NaN	NaN	NaN	NaN	NaN	1.0	113.0	0.0	1.4

3075 rows × 10 columns

```
In [28]: merged= pd.merge(df1, df2, on='cp', how='inner')
print("Integrated dataset shape:", merged.shape)
merged
```

Integrated dataset shape: (361483, 6)

Out[28]:

	age	sex	cp	trestbps	chol	fbs
0	52	1	0	125	212	0
1	52	1	0	140	203	1
2	52	1	0	145	174	0
3	52	1	0	148	203	0
4	52	1	0	138	294	1
...	...	...	...	...	...	...
361478	54	1	0	112	204	0
361479	54	1	0	125	258	0
361480	54	1	0	110	275	0
361481	54	1	0	110	254	0
361482	54	1	0	120	188	0

361483 rows × 6 columns

```
In [34]: sort_values = df.sort_values('cp', ascending=True)
sort_values
```

```
Out[34]:
```

	age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	slope	ca	thal
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3
579	58	1	0	146	218	0	1	105	0	2.0	1	1	3
583	43	1	0	120	177	0	0	120	1	2.5	1	0	3
584	55	1	0	132	353	0	1	132	1	1.2	1	1	3
587	59	1	0	164	176	1	0	90	0	1.0	1	2	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...
959	69	0	3	140	239	0	1	151	0	1.8	2	2	2
104	52	1	3	118	186	0	0	190	0	0.0	1	0	1
714	51	1	3	125	213	0	0	125	1	1.4	2	1	2
627	38	1	3	120	231	0	1	182	1	3.8	1	0	3
664	66	0	3	150	226	0	1	114	0	2.6	0	0	2

1025 rows × 14 columns



```
In [36]: sort_values = df.sort_values('age', ascending=True)
sort_values
```

Out[36]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
60	29	1	1	130	204	0	0	202	0	0.0	2	0	2
118	29	1	1	130	204	0	0	202	0	0.0	2	0	2
668	29	1	1	130	204	0	0	202	0	0.0	2	0	2
64	29	1	1	130	204	0	0	202	0	0.0	2	0	2
572	34	1	3	118	182	0	0	174	0	0.0	2	0	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
99	76	0	2	140	197	0	2	116	0	1.1	1	0	2
965	76	0	2	140	197	0	2	116	0	1.1	1	0	2
160	77	1	0	125	304	0	0	162	1	0.0	2	3	2
162	77	1	0	125	304	0	0	162	1	0.0	2	3	2
387	77	1	0	125	304	0	0	162	1	0.0	2	3	2

1025 rows × 14 columns

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▶

In [38]:

df.transpose()

Out[38]:

	0	1	2	3	4	5	6	7	8	9	...	1015	101
age	52.0	53.0	70.0	61.0	62.0	58.0	58.0	55.0	46.0	54.0	...	58.0	65.
sex	1.0	1.0	1.0	1.0	0.0	0.0	1.0	1.0	1.0	1.0	...	1.0	1.
cp	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	3.
trestbps	125.0	140.0	145.0	148.0	138.0	100.0	114.0	160.0	120.0	122.0	...	128.0	138.
chol	212.0	203.0	174.0	203.0	294.0	248.0	318.0	289.0	249.0	286.0	...	216.0	282.
fbs	0.0	1.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	...	0.0	1.
restecg	1.0	0.0	1.0	1.0	1.0	0.0	2.0	0.0	0.0	0.0	...	0.0	0.
thalach	168.0	155.0	125.0	161.0	106.0	122.0	140.0	145.0	144.0	116.0	...	131.0	174.
exang	0.0	1.0	1.0	0.0	0.0	0.0	0.0	1.0	0.0	1.0	...	1.0	0.
oldpeak	1.0	3.1	2.6	0.0	1.9	1.0	4.4	0.8	0.8	3.2	...	2.2	1.
slope	2.0	0.0	0.0	2.0	1.0	1.0	0.0	1.0	2.0	1.0	...	1.0	1.
ca	2.0	0.0	0.0	1.0	3.0	0.0	3.0	1.0	0.0	2.0	...	3.0	1.
thal	3.0	3.0	3.0	3.0	2.0	2.0	1.0	3.0	3.0	2.0	...	3.0	2.
target	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.

14 rows × 1025 columns



```
In [41]: # Melt (reshape) the data - from wide to Long format
df_resaped = pd.melt(df, id_vars=['age', 'sex'], var_name='measurement', value_name='value')

# Display reshaped data
print(df_resaped.head())
```

```
   age  sex measurement  value
0   52    1          cp    0.0
1   53    1          cp    0.0
2   70    1          cp    0.0
3   61    1          cp    0.0
4   62    0          cp    0.0
```

```
In [ ]: #reshape
pivot_table=pd.pivot_table(df,index=['age','sex']
```